

Advanced Chromatographic Techniques

Science

May, 2026

Advanced Chromatographic Techniques (A11567)

Short Title: Adv. Chromatogr. Techniques
Faculty Science and Computing
Department Science
Cluster Analytical Science
Author KGRENNAN
Credits 10

Level: Postgraduate

Description of Module / Aims

This programme will provide the student with an advanced understanding of the theory, practices and instrumentation associated with advanced in separation science as well as interfacing of hyphenated techniques involving mass-spectrometry such as LC-MS and GC-MS. The learner will be exposed to state-of-the-art developments in associated instrumentation, as well as issues to deal with the separation of samples in complex matrices. The necessary background will be given which will allow the learner to select an appropriate separation technique for a given analysis. The learner will be exposed to the application of advanced separation techniques in a wide range of environments focusing mainly, but not exclusively, on applications in the pharmaceutical and biopharmaceutical industries.

Programmes

SCIE-0050	Certificate in Advanced Chromatographic Techniques (WD_SCHRM_MA)	5 / 0 / M
SCIE-0050	MSc in Analytical Science with Quality Management (WD_SANSC_R)	1 / 0 / E
SCIE-0050	PG Dip in Analytical Science with Quality Management (WD_SANSC_G)	1 / 0 / E

Indicative Content

- Advanced options & state-of-the-art developments in Liquid Chromatography (LC), e.g. stationary phase & column technology, instrumentation, micro- and nano-column LC, separation on chips, ultra-high pressure LC, high-performance TLC, 2D-LC, LC-MS interfacing and mobile phase considerations
- Advanced options & state-of-the-art developments in Gas Chromatography (GC), e.g. instrumentation, sensitivity enhancement, thermally labile samples, analytes in complex matrices, detection systems, efficiency increases and automation, 2D-GC and GC-MS interfacing
- Issues relating to sample pre-treatment in complex matrices, e.g. solid phase extraction (SPE) techniques, solid phase micro-extraction (SPME), supercritical fluids, fibre and disk extraction, programmed injection methodologies in GC. Analysis of small molecules
- Advanced options & state-of-the-art developments in ion chromatography (IC), e.g. instrumentation, anion & cation chromatography separations, speciation of metals
- Mass spectroscopy (MS) detection techniques coupled to chromatographic separations: LC-MS & GC-MS - in-depth theoretical treatment & applications
- Reverse-phase bioseparation, e.g. peptide mapping, sample preparation in peptide mapping and types of stationary phases applicable for LC separation of large biomolecules
- Chiral analysis of a range of materials, particularly pharmaceuticals, using LC, GC and CE and chiroptical analysis
- Laboratory practicals demonstrating ability to operate instrumentation associated with the above techniques

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise the principles, practices and instrumentation associated with advanced chromatography, as well as interfacing of hyphenated techniques involving mass-spectrometry such as LC-MS and GC-MS.
2. Select an appropriate separation technique for a given analysis in a range of industrial environments.
3. Solve sample preparation issues concerned with GC, HPLC, ion chromatography and reverse-phase bioseparations.
4. Solve method development issues concerned with GC, HPLC, ion chromatography and reverse-phase bioseparations.
5. Set up, calibrate and qualify associated laboratory instrumentation for pharmaceutical and biopharmaceutical sample analyses.
6. Appraise separation science literature and identify future trends in this area.

Learning and Teaching Methods

- Lectures.
- Practicals.
- E-learning.
- IT practicals.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Practical</i>	50%	3,4,5
<i>In-Class Assessment</i>	35%	1,2
<i>Assignment</i>	15%	6

Assessment Criteria

- <40%:** The learner will demonstrate little or no understanding of the fundamentals of the learning outcomes. Will display a lack of competence with regards to practical and resulting data analysis skills.
- 40%-59%:** The learner will demonstrate limited understanding of the fundamentals of the learning outcomes. Will display some ability with regards to associated practical and resulting data analysis skills.
- 60%-69%:** As well as the above, the learner will demonstrate reasonable understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regards to associated practical and resulting data analysis skills.
- 70%-100%:** All of the above to excellent levels. In addition, the learner will demonstrate comprehensive knowledge and understanding of all learning outcomes. Will display advanced competence and independence with regards to associated practical and resulting data analysis skills.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Practical		24
Independent Learning		222

Essential Material

- "Advanced Chromatographic Techniques module Moodle page." <http://moodle.wit.ie>
- "WIT library databases." <http://library.wit.ie/Resources/databases>

Requested Resources

- Room Type: Chemistry Lab
- Room Type: Computer Lab